INVERSE OF THE RESTRICTED TANGENT

Profile of arctangent

$y = \arctan(x)$

ℝ DOMAIN · 2 HORIZONTAL ASYMPTOTES



KEY POINTS

1 THROUGH
$(0, 0)$

2 AT X = ±1
$y = \pm\pi/4$

3 ASYMPTOTES
$y = \pm\pi/2$

GLOBAL

DOMAIN
$(-\infty, \infty)$
tangent's range is unbounded

RANGE
$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
open: the ends are never reached

SYMMETRY
odd: $\arctan(-x) = -\arctan x$

FAR ENDS
$\arctan x \to \pm\frac{\pi}{2}$ as $x \to \pm\infty$